

1.

a. State

A state in this puzzle is defined by the arrangement of coins in the seven cells. Each state can be represented by a string of seven characters where each character can be 'B' for a blue coin, 'R' for a red coin, or '0' for an empty cell. E.g, State A can be represented as "BRBR0RB" and State B as "RRR0BBB".

b. Goal State

The goal state is the desired arrangement of coins where blue coins occupy cells 1, 2, and 3, and red coins occupy cells 5, 6, and 7. This can be represented as "BBBRRR" or more formally as an array where the first three positions are 'B' and the last three positions are 'R', with the fourth position being '0' (empty).

c. Moves (Actions)

Moves or actions in this puzzle involve either:

1. Moving a coin to an adjacent empty cell if that cell is directly next to it.
2. Jumping a coin over an adjacent occupied cell into an empty cell two cells away.

E.g, from State A ("BRBR0RB"):

- A red coin in cell 4 can move to cell 5 if cell 5 is empty.
- A blue coin in cell 3 can jump over the red coin in cell 4 into cell 5 if cell 5 is empty.

2.

0
B

